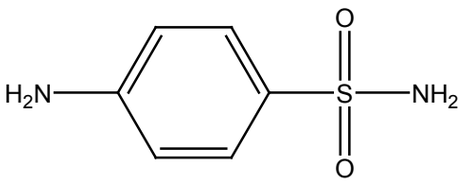
Enzymes as drug targets:

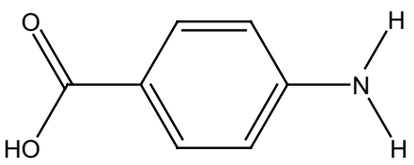
In all living systems, the biochemical reactions are catalysed by enzymes. Hence, these enzyme actions are highly essential for the normal functioning of the system. If their normal enzyme activity is inhibited, then the system will be affected. This principle is usually applied to kill many pathogens.

We have already learnt that in enzyme catalysed reactions, the substrate molecule binds to the active site of the enzyme by means of the weak interaction such as hydrogen bonding, van der Waals force etc… between the amino acids present in the active site and the substrate.

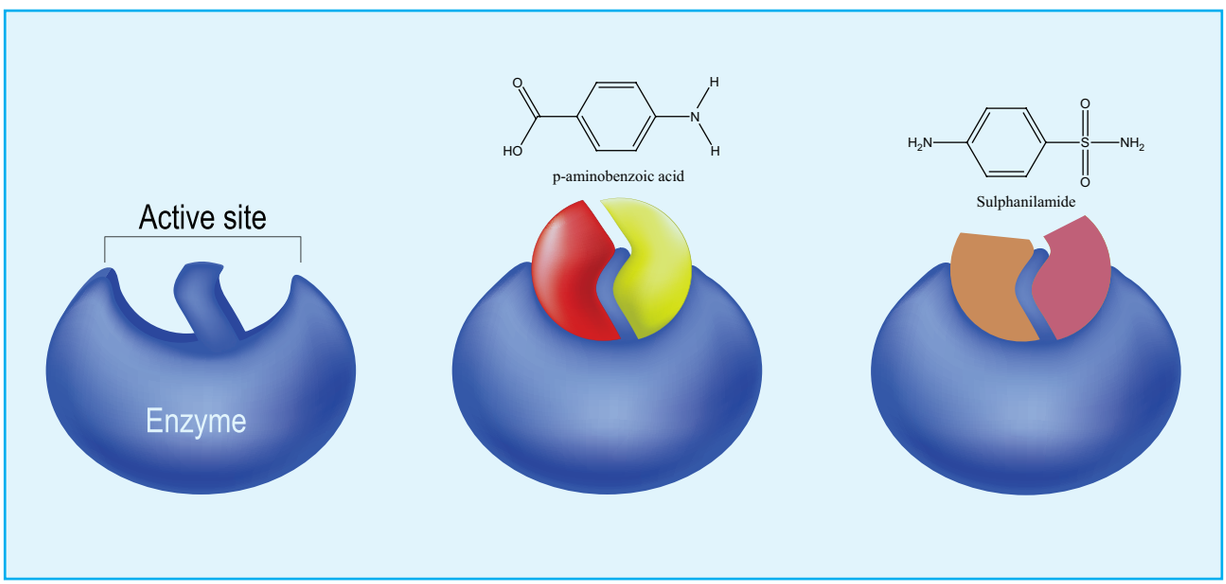
When a drug molecule that has a similar geometry (shape)as the substrate is administered, it can also bind to the enzyme and inhibit its activity.In other words, the drug acts as an inhibitor to the enzyme catalyst. These type of inhibitors are often called competitive inhibitors. For example the antibiotic sulphanilamide, which is structurally similar to *p*-aminobenzoic acid (PABA) inhibits the bacterial growth. Many bacteria need PABA in order to produce an important coenzyme, folic acid. When the antibiotic sulphanilamide is administered, it acts as a competitive inhibitor to the enzyme dihydropteroate synthase (DHPS) in the biosynthetic pathway of converting PABA into folic acid in the bacteria. It leads to the folic acid deficiency which retards the growth of the bacteria and can eventually kill them.



Sulphanilamide



p-aminobenzoic acid



In certain enzymes, the inhibitor molecule binds to a different binding site, which is commonly referred to as allosteric site, and causes a change in its active site geometry (shape). As a result, the substrate cannot bind to the enzyme. This type of inhibitors are called allosteric inhibitors.

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